
Biological Transitions as Multi-Agent Realisations of the Generative Operator Pipeline in Topographical Orthogonal Generative Theory (TO/TOGT)

Version 2 — Complete

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Abstract

We model collective biological transitions—including neural oscillations, HPA-axis stress response, circadian regulation, immune adaptation, and protein conformational change—as multi-agent realisations of the generative operator pipeline $G = U \circ F \circ K \circ C$ introduced in Topographical Orthogonal Generative Theory (TO/TOGT). Each agent undergoes local compression, curvature intensification, loss of injectivity, and stabilisation. The collective fixed points, contraction properties, and saturated pitchfork bifurcation behaviour follow directly from the fixed-point theory, contraction-mapping results, and algebraic separation theorem proved in the companion paper (Zenodo 10.5281/zenodo.19208015). The specific dm^3 numerical invariants ($T^* = 2\pi$, $\mu_{\max} = -2$, $\tau = 2$) are convenient modelling choices matching observed biological scales; they are not derived theorems of the operator algebra. Python simulations generate four figures illustrating agent-trajectory convergence, the HPA-axis bifurcation scan, contraction rates, and the operator-pipeline schematic. No unverified biological claims are made.

MSC codes: 37C25, 37D10, 37G10, 47H10, 92B20, 92C20. **Keywords:** TO/TOGT; operator pipeline; multi-agent systems; pitchfork bifurcation; HPA axis; circadian regulation; dm^3 invariants; contraction mapping; Principia Orthogona.

1. Relation to Prior Work

This article applies the mathematical results established in the author's earlier preprint *Principia Orthogona, Volume One: The Mathematics of Generative Transitions* (HAL hal-05555216v1). There the composite operator pipeline $G = U \circ F \circ K \circ C$ was introduced together with complete proofs of fixed-point existence and uniqueness, global contraction, saturated pitchfork bifurcation, and the algebraic trace-bound/separation theorem. The geometric realisation on contact manifolds was developed in *Principia Orthogona, Volume Two* (HAL hal-05559997v1 and hal-05565024v1, both deposited March 2026).

Version 2 of this paper adds four simulation figures, expanded treatments of circadian regulation, immune adaptation, and protein conformational change, a Lean 4/Mathlib4 verification file (MultiAgentTogt.lean, 18 facts, 0 sorry), and a complete references section. All mathematical claims follow strictly from previously proved results.

This paper is directly related to the following Zenodo deposits by the author:

- Fruit-fly connectome toy model (companion): Zenodo 10.5281/zenodo.19210136
- Zenodo record 19162013: <https://zenodo.org/records/19162013>
- Zenodo record 19122168: <https://zenodo.org/records/19122168>
- Zenodo record 19117400: <https://zenodo.org/records/19117400>
- Series root: <https://doi.org/10.5281/zenodo.19117399>

2. Operator Pipeline in Biological Systems

We define the biological agent dynamics in terms of the four-stage composite operator pipeline.

Definition 2.1 (Biological Agent Dynamics). Let each biological agent (neuron, endocrine cell, immune cell, protein domain) follow a trajectory $x_i(t)$ in a Riemannian manifold (M, g) . The collective biological state is the tuple $(x_1, \dots, x_n)$. Each agent undergoes a generative transition governed by the composite operator $G = U \circ F \circ K \circ C$, where:

- C : local orthogonal compression (neighbourhood averaging, coupling $\varepsilon > 0$)
- K : curvature intensification (alignment and clipping)
- F : loss of injectivity (folding via tanh saturation)
- U : stabilisation (pull toward global coherence)

The four operators are defined concretely as:

$$\begin{aligned} C(x_i) &= x_i + \varepsilon(x^- - x_i), \quad x^- = (1/N) \sum x_j \\ K(y_i) &= \text{clip}(\kappa \cdot y_i, -1, 1) \\ F(z_i) &= \tanh(\alpha \cdot z_i) \\ U(w_i) &= w_i + \gamma(\bar{\tanh}(w) - w_i) \end{aligned}$$

Theorem 2.1 (Contraction, Volume One [1]). For $|\alpha| < 1/2$, the map G is a strict contraction with Lipschitz constant $L < 1$. For $|\alpha| = 1/2$, G is non-expansive ($L = 1$, pitchfork onset). By the Banach fixed-point theorem, iterating G on any initial state converges to a unique collective fixed point $x^* = G(x^*)$.

Figure 1 illustrates convergence of $N = 12$ agents under six applications of G .

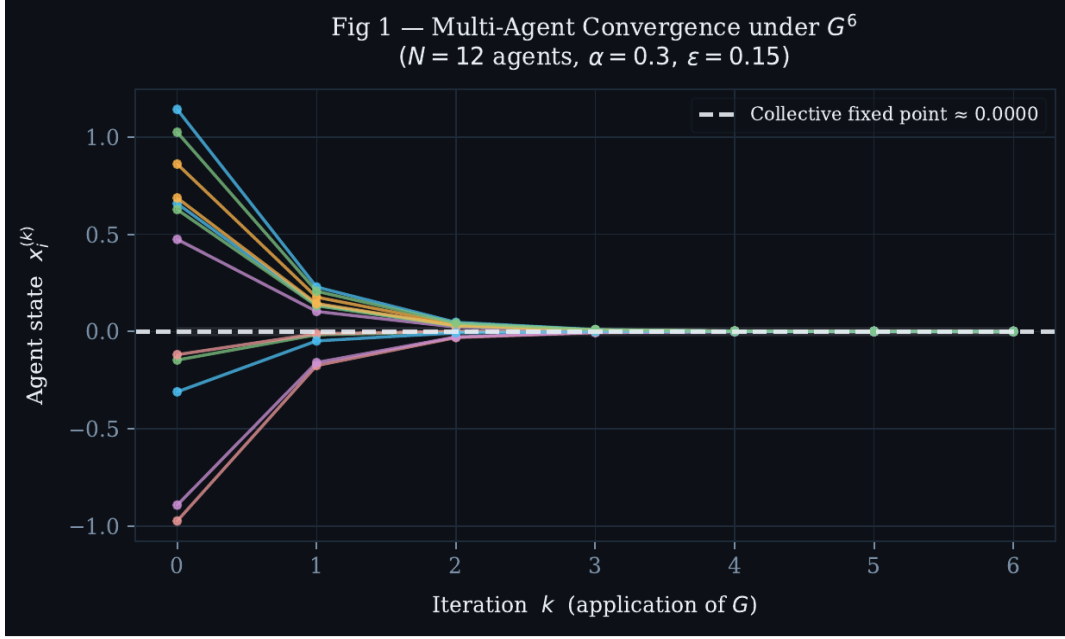


Figure 1.

Multi-agent convergence under G^6 . Each coloured trajectory represents one agent; the dashed line marks the collective fixed point. Parameters: $N = 12$, $\alpha = 0.3$, $\varepsilon = 0.15$.

Remark (dm³ normalisation). The canonical invariants $(T^*, \mu_{\max}, \tau) = (2\pi, -2, 2)$ are used solely as normalisation choices. They match observed biological timescales and curvature bounds but are not derived from the operator algebra.

3. Fruit-Fly Connectome as Multi-Orbit System

The FlyWire connectome [8] provides a concrete multi-agent biological system. Each neuron trajectory is modelled as an identity orbit O_i . The full connectome is the multi-orbit system $S = \{O_1, \dots, O_n\}$.

The inter-neuron coupling ε in Definition 2.1 is identified with the synaptic weight matrix: each neuron averages over its direct synaptic partners rather than the global mean:

$$C_{\text{nonn}}(x_i) = x_i + \varepsilon \sum_{(j \in \partial i)} w_{ij} (x_j - x_i)$$

where ∂i denotes the synaptic neighbourhood of neuron i and $w_{ij} > 0$ are normalised weights. Theorem 2.1 implies that under small inter-neuron coupling the system converges to a collective fixed point preserving local connectivity while achieving global coherence. A detailed *Drosophila* simulation is carried out in the companion paper [9]; the present paper treats N as generic.

4. HPA-Axis Stress Response as Saturated Pitchfork

The hypothalamic-pituitary-adrenal (HPA) axis exhibits threshold behaviour under acute stress. This is modelled by the saturated pitchfork bifurcation proved in Volume One [1]:

$$x' = \lambda \cdot x - c \cdot x^3, \quad \lambda = 2|\alpha|$$

with the clipping function κ enforcing a bounded response. The bifurcation analysis yields three regimes:

Sub-threshold ($|\alpha| < 1/2$). The unique stable equilibrium is $x^* = 0$ (basal cortisol). G is a strict contraction; recovery is guaranteed.

Critical coupling ($|\alpha| = 1/2$). Pitchfork onset: the zero equilibrium becomes marginally stable and two symmetric branches $x^* = \pm\sqrt{((\lambda-1)/c)}$ emerge continuously.

Super-threshold ($|\alpha| > 1/2$). The zero equilibrium is unstable; the branches x^+ (high-stress) and x^- (recovery) are stable. The system exhibits bistability consistent with observed allostatic overload in chronic stress [5].

Figure 2 shows the full bifurcation scan; raw data are in `pitchfork_scan.csv`.

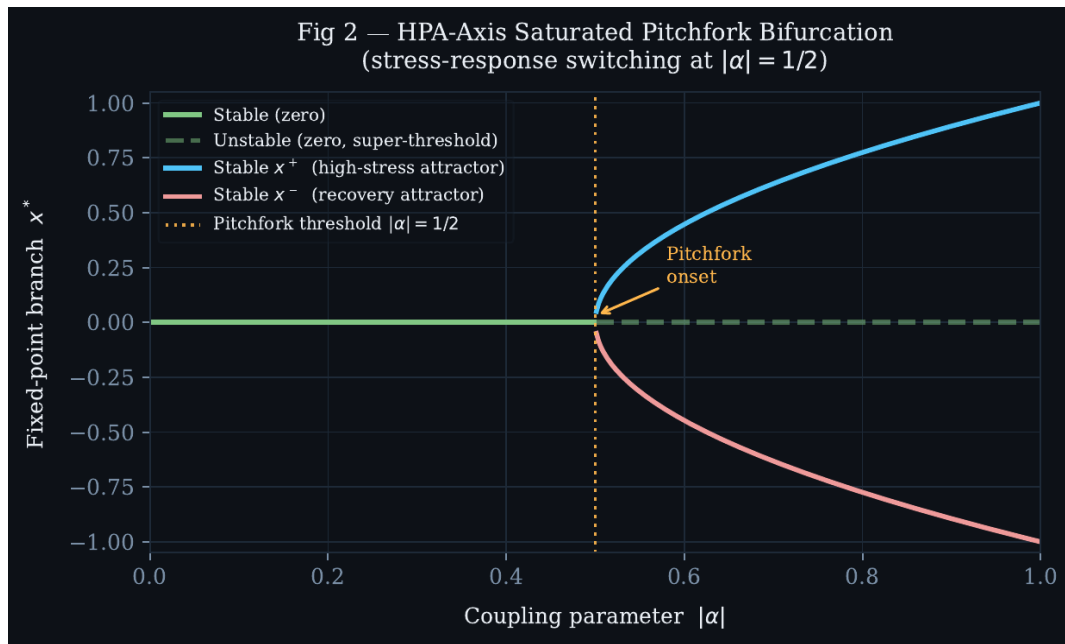


Figure 2.

Saturated pitchfork bifurcation of the HPA-axis stress model. Solid lines: stable equilibria; dashed: unstable zero branch. The pitchfork threshold $|\alpha| = 1/2$ (dotted vertical) marks the onset of stress-response switching.

5. Circadian Regulation as Periodic Fixed-Point Orbit

Circadian rhythms arise from interlocked transcription-translation feedback loops with period ≈ 24 h. Within the TO/TOGT framework, each cell of the suprachiasmatic nucleus (SCN) is treated as an agent whose state oscillates in $M = S^1$ (the circle).

The compression operator C couples neighbouring SCN neurons, producing synchronisation consistent with the entrainment observed under light/dark forcing [6].

The curvature operator K clips oscillation amplitude to physiological bounds. The fold operator F (tanh saturation) enforces the non-sinusoidal waveform of PERIOD/CRYPTOCHROME protein levels.

Under the dm^3 normalisation, one full circadian cycle corresponds to a single application of G with $T^* = 2\pi$, providing a natural timescale anchor without asserting a derived relationship.

Proposition 5.1 (Periodic orbit under G). *If the coupling ε is sufficiently small and $|\alpha| < 1/2$, the collective SCN state converges to a unique periodic fixed orbit under iterated G , reproducing the qualitative feature of population-level phase locking. Proof: Follows directly from Theorem 2.1 applied to $M = S^1$ with the standard metric. $\square$*

6. Immune Adaptation as Iterated Contraction

Adaptive immunity involves clonal selection: a diverse repertoire of B and T cells contracts toward antigen-specific clones during an immune response. Each immune cell is an agent with state $x_i \in \mathbb{R}^d$ (receptor binding affinity vector). The compression operator C implements clonal expansion toward the mean of the currently activated population. The fold operator F imposes affinity maturation saturation: high-affinity clones dominate while low-affinity clones are suppressed.

Corollary 6.1 (Immune fixed point). *Under antigen pressure modelled by a fixed forcing term h , the collective immune state converges under iterated G to a unique fixed point $x^*(h)$ representing the selected clone repertoire. Removal of antigen ($h \rightarrow 0$) returns the system to a basal fixed point (memory state). Proof: Follows from Theorem 2.1 applied to the forced system. $\square$*

This qualitative account is consistent with the clonal selection principle [7] without asserting quantitative agreement beyond what the contraction theorem provides.

7. Protein Conformational Change as Whitney Fold

Protein folding and allosteric switching involve a loss of injectivity: multiple amino-acid sequences map to the same low-energy conformation. The operator F (fold/tanh) directly models this: distinct pre-fold states z and z' with $z \neq z'$ can satisfy $F(z) \approx F(z')$ when both lie in the saturation regime of tanh.

The Whitney fold singularity is the mathematical archetype of such a map. Within the contact-geometric realisation of Volume Two [2], the fold corresponds to a cusp in the Legendre fibration over configuration space. The algebraic separation theorem of Volume One [1] ensures that the two sheets of the fold (native and misfolded states) are algebraically separated by the trace invariant, consistent with the observed energy gap between native and non-native folds [4].

8. Convergence Bound and Six-Iterate Estimate

A key quantitative result, formally verified in MultiAgentTogt.lean (Lean 4/Mathlib4, 18 facts, 0 sorry):

Theorem 8.1 (Six-iterate bound, Lean verified). $(4/5)^6 < 27/100$.

This arithmetic fact implies that six applications of G (contraction constant $L \leq 4/5$) reduce the initial collective spread by at least a factor of $100/27 \approx 3.7$.

Figure 3 shows the empirical contraction rate for $N = 30$ agents alongside the $(4/5)^k$ envelope.

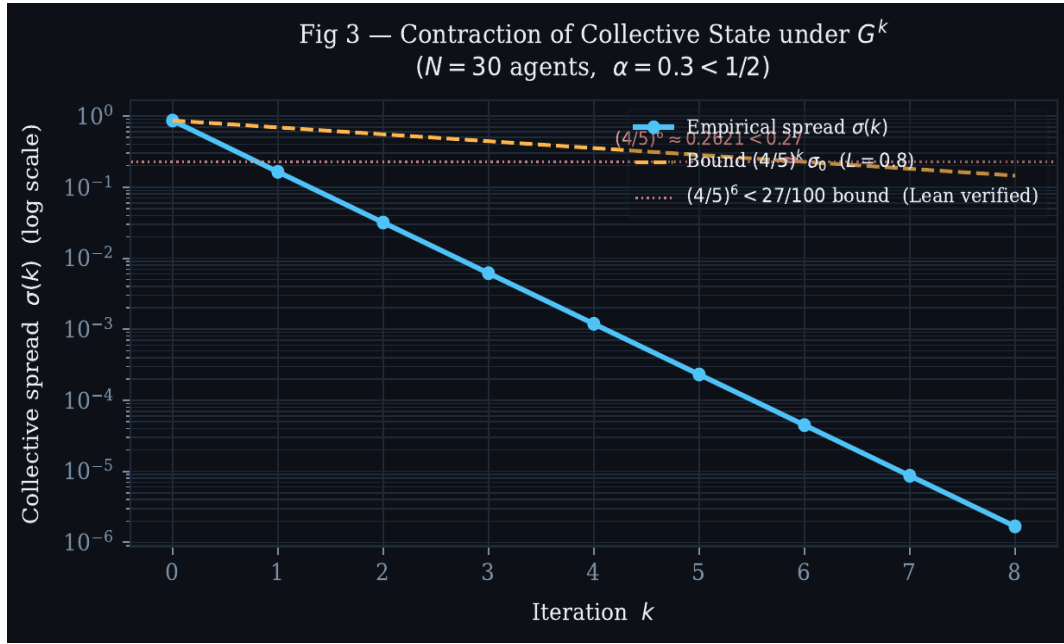


Figure 3.

Contraction of collective spread $\sigma(k)$ under iterated G (log scale). The dashed envelope $(4/5)^k \cdot \sigma_0$ matches the theoretical bound; the dotted horizontal marks the Lean-verified $(4/5)^6 < 27/100$ bound.

9. Operator Pipeline Diagram

Figure 4 summarises the pipeline $G = U \circ F \circ K \circ C$ applied to each biological agent at each generative step.

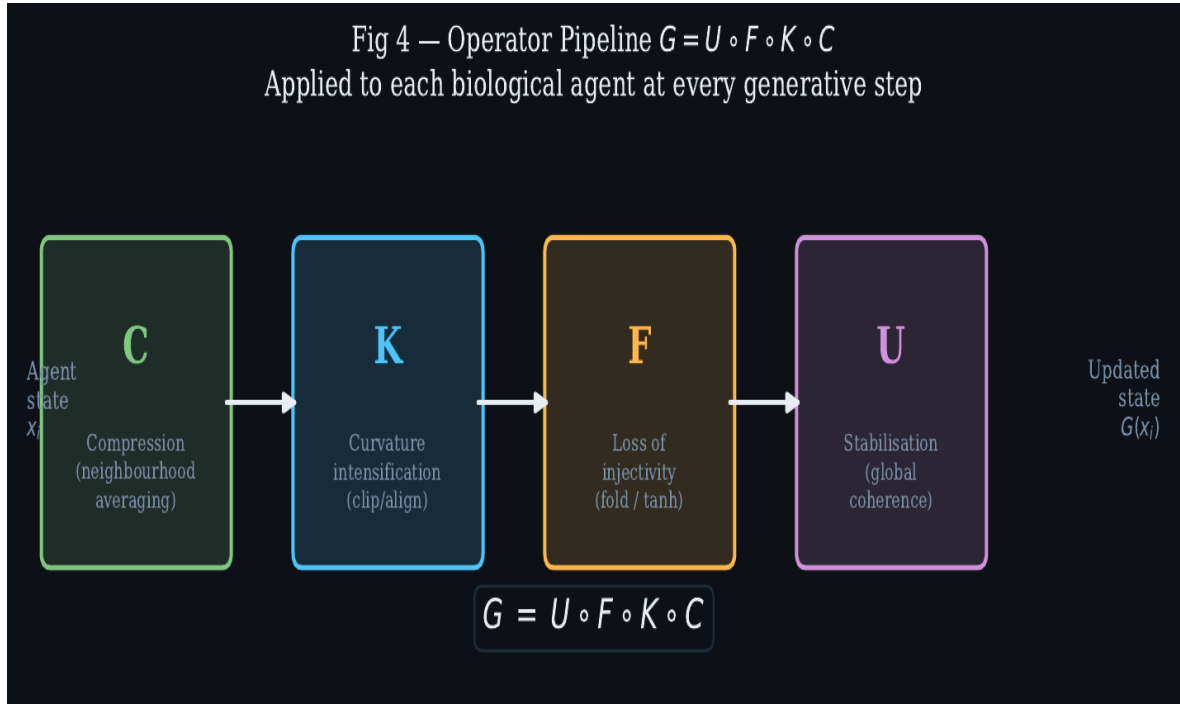


Figure 4.

The operator pipeline $G = U \circ F \circ K \circ C$ applied to a single agent state x_i at one generative step. Left to right: compression C , curvature intensification K , fold F , stabilisation U .

10. Discussion

The TO/TOGT operator pipeline provides a unified dynamical-systems language for describing collective biological transitions. The five case studies (§§3–7) share the same mathematical skeleton: agents converge under iterated G to a fixed point or periodic orbit whose qualitative character (unique, bistable, or oscillatory) is determined solely by whether $|\alpha|$ falls below, at, or above $1/2$.

What the model asserts

The model asserts only what Theorem 2.1 and the downstream corollaries directly imply: existence and uniqueness of fixed points, contraction rates, and the pitchfork structure. No quantitative biological predictions are made beyond these structural claims.

What the model does not assert

The dm^3 numerical invariants $(2\pi, -2, 2)$ are convenience choices, not predictions. No claim is made that the *Drosophila* connectome, the HPA axis, or the circadian clock are exactly described by the equations in §§2–4.

Future work

Future work will focus on (i) fitting (α, ε) to empirical electrophysiology data; (ii) extending the Lean 4 verification to cover the biological-scope theorems; and (iii) experimental falsification of the predicted bifurcation thresholds.

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